COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212 - Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	09/24/2026	**DATE SUBMITTED:**	09/24/2026
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Midterm Skills Exam

1. PROCEDURE OUTPUT

Sample output (students are allowed to add more steps as needed.)
Task 1: Prepare the Server

```
  GNU nano 7.2                                                  inventc
all:
  vars:
    ansible_user: charles-vence-espinas

  children:
    Ubuntu_servers:
      hosts:
        server1:
          ansible_host: 192.168.199.152

    CentOS_servers:
      hosts:
        server2:
          ansible_host: 192.168.199.7

    linux_servers:
      children:
        Ubuntu_servers:
        CentOS_servers:
```

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab/Midterm_Skills_Exam$ ansible all -i inventor
.yaml -m ping
server1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
server2 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
```

- I setup the server and ping it to test if its reachable

```yaml
---
- hosts: all
  become: true
  tasks:

    - name: Update repository in ubuntu
      ansible.builtin.apt:
        upgrade: dist
        update_cache: yes
      when: ansible_distribution == "Ubuntu"

    - name: Update repository in centos
      ansible.builtin.yum:
        upgrade: dist
        update_cache: yes
      when: ansible_distribution == "CentOS"

    - name: installing  important utilities in prometheus in ubuntu
      ansible.builtin.apt:
        name:
          - wget
          - tar
          - curl
        state: present
      when: ansible_distribution == "Ubuntu"
```

```yaml
    - name: installing important utilities in prometheus in centos
      ansible.builtin.yum:
        name:
          - wget
          - tar
          - curl
        state: present
      when: ansible_distribution == "CentOS"
```

- I install the utilities that I will be needed later in this activity.

Task 2: Create the Prometheus User

```yaml
    - name: Create a group
      ansible.builtin.group:
        name: prometheus
        state: present


    - name: Create a user and assign it to a group
      user:
        name: prometheus
        groups: prometheus
        append: yes
```

- I create the group ere using a .builtin.group with the name prometheus and also name of the user which name is prometheus.

Task 3: Create Prometheus Directories

```yaml
- name: Create Prometheus Directory
  ansible.builtin.file:
    path: /etc/prometheus
    state: directory
    mode: '755'

- name: Create a Prometheus Libraries
  ansible.builtin.file:
    path: /var/lib/prometheus
    state: directory
    mode: '755'
```

- I create a directory for /etc/prometheus and /var/lib/prometheus in both servers and I set the mode to 755 so the owner has read, write and execute and the groups and to others read and execute.

Task 4: Download Prometheus

```yaml
- name: Download Prometheus
  ansible.builtin.get_url:
      url: https://github.com/prometheus/prometheus/releases/download/v2.55.1/prometheus-2.55.1.linux-amd64.tar.gz
      dest: /tmp
      mode: '755'

- name: Extract Prometheus
  ansible.builtin.unarchive:
    src: /tmp/prometheus-2.55.1.linux-amd64.tar.gz
    dest: /tmp
    remote_src: yes
    mode: '755'
    owner: prometheus
    group: prometheus
```

- I download the prometheus from github using builtin.get_url from the github and then I extract it using builtin.unarchive.

Task 5: Install the Prometheus Binaries

```yaml
- name: Install Prometheus binaries (prometheus)
  ansible.builtin.copy:
    src: /tmp/prometheus-2.55.1.linux-amd64/prometheus
    dest: /usr/local/bin/
    remote_src: yes
    mode: '755'
    owner: prometheus
    group: prometheus

- name: Install Prometheus Binaries (promtool)
  ansible.builtin.copy:
    src: /tmp/prometheus-2.55.1.linux-amd64/promtool
    dest: /usr/local/bin/
    mode: '755'
    remote_src: yes
    owner: prometheus
    group: prometheus
```

Task 6: Install the Prometheus Configuration Files

```
- name: Install prometheus configuration file
  ansible.builtin.copy:
    src: /tmp/prometheus-2.55.1.linux-amd64/prometheus.yml
    dest: /etc/prometheus
    remote_src: yes
    owner: prometheus
    group: prometheus

- name: Install prometheus console
  ansible.builtin.copy:
    src: /tmp/prometheus-2.55.1.linux-amd64/consoles
    dest: /etc/prometheus
    remote_src: yes
    owner: prometheus
    group: prometheus

- name: Install prometheus console libraries
  ansible.builtin.copy:
    src: /tmp/prometheus-2.55.1.linux-amd64/console_libraries
    dest: /etc/prometheus
    remote_src: yes
    owner: prometheus
    group: prometheus
```

Task 7: Validate the Configuration
Task 8: Create the systemd Service
Task 9: Start and Enable Prometheus
Task 10: Verify Prometheus
Task 11: Access Prometheus Through a Browser

```
charles-vence-espinas@workstation:~/CPE212_CharlesVenceEspinas/ansible-lab/Midterm_Skills_Exam$ ansible-playbook -i inv
ntory.yaml installing_prometheus.yml --ask-become-pass
BECOME password:

PLAY [all] ********************************************************************************************************

TASK [Gathering Facts] ********************************************************************************************
ok: [server2]
ok: [server1]

TASK [installing  important utilities in prometheus in ubuntu] ****************************************************
skipping: [server2]
ok: [server1]

TASK [installing important utilities in prometheus in centos] *****************************************************
skipping: [server1]
ok: [server2]

TASK [Create a group] *********************************************************************************************
ok: [server2]
ok: [server1]

TASK [Create a user and assign it to a group] *********************************************************************
ok: [server2]
ok: [server1]

TASK [Create Prometheus Directory] ********************************************************************************
ok: [server2]
ok: [server1]
```

```
TASK [Create a Prometheus Libraries] ******************************************************************************
ok: [server1]
ok: [server2]

TASK [Download Prometheus] ****************************************************************************************
changed: [server2]
changed: [server1]

TASK [Extract Prometheus] *****************************************************************************************
changed: [server2]
changed: [server1]

PLAY RECAP ********************************************************************************************************
server1                    : ok=8    changed=2    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
server2                    : ok=8    changed=2    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
```

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2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assis concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, a completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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